

THEJUS MAHAJAN PH.D.

Bioinformatics & Data Scientist

+49-15259697629

thejusmahajan.github.io

@ thejus.mahajan@uni-hamburg.de

Hamburg, Germany

*30.09.1992 (Talikulam, India)

Married



PROFILE

- **Focus:** Large-scale data analysis, bioinformatics pipeline development, and statistical modeling.
- **Core Stack:** R (Bioconductor, Tidymodels), Python (Pandas, JAX, Scikit-learn), Nextflow (DSL2), BASH.
- **Current Objective:** Applying reproducible computational workflows and AI methodologies to advance translational collaborations in biomedicine.

EXPERIENCE

Internship – Bioinformatics & Data Pipeline Engineering

HealthTwist GmbH

02/2026 - 04/2026

Berlin, Germany

Clinical data research and development under Dr. Andreas Busjahn

- Refactored a monolithic 1,834-line R data pipeline (tidyverse) for Germany's interventional radiology quality registry (DeGIR); 143,000+ patient records from ~300 clinics. Reduced to 1,348 lines (–26.5%) with byte-identical output verified via `identical()`.
- Externalised 257 hard-coded correction rules into 8 configuration CSV files, enabling non-programmer editing of business rules.
- Replaced 357 deprecated function calls across 4 pipeline files, modernising the codebase to current tidyverse standards.
- Created the `degirtools` R package (9 exported functions, roxygen2 docs, devtools::check() passing) by extracting reusable logic from a 767-line helper file.
- Discovered 2 pre-existing bugs through systematic code analysis; documented both and preserved original behaviour for supervisor review – demonstrating production-code discipline.
- Built an interactive R Shiny dashboard (6 modules: overview, interventions, complications, radiation doses, success rates, about) using plotly, DT, and bslib. Deployed to shinyapps.io with GDPR-safe synthetic data.
- Wrote Tidymodels teaching materials (Quarto/GitHub) replacing the legacy caret framework for ML workflows in R.

Further Training – Bioinformatics and Biostatistics

CQ Beratung + Bildung GmbH

08/2025 - 02/2026

Berlin, Germany

Application-related Bioinformatics and Biostatistics

- Programming methodology: Shell (BASH), Python (object-oriented), R.
- Bioinformatics databases (NCBI, Biopython, SQL) and sequence analysis.
- NGS analysis pipelines (Nextflow, Galaxy); structural bioinformatics.
- Clinical biostatistics (R/Bioconductor, ANOVA, PCA, HCA, survival analysis); GDPR/BDSG compliance.
- Capstone: Built a reproducible Nextflow (DSL2) pipeline for Hepatitis Delta Virus sequence analysis – automated NCBI download, MAFFT alignment, and trimAI trimming with Conda-managed dependencies (🔗 GitHub).

Guest Scientist

Helmholtz-Zentrum Hereon, Ökosystemmodellierung

05/2025 - 10/2025

Geesthacht, Germany

Computational Biologist & Systems Modeler

- Spearheaded research modeling evolutionary processes in marine ecosystems.
- Applied advanced computational techniques to simulate long-term environmental impacts.

STRENGTHS



Data Pipeline Engineering

Refactored a production data pipeline (1,834 lines, 143K records, ~300 clinics) with 26.5% code reduction. Built `degirtools` R package. Byte-level verification after every change via `identical()`.



Complex Problem-Solving

Resolved a critical data-loss error by creating a Fortran simulation to correct for a misplaced particle detector, recalculating results using multi-sensor data.



Bioinformatics & Biostatistics

Bridged physical science expertise with modern biological workflows. Skilled in complex multidimensional array operations, JAX/GPU acceleration, sequence alignment, and multivariate statistical modeling using R (Bioconductor) and Python.

TECHNICAL SKILLS

Programming

Python (JAX, Pandas, NumPy, Scikit-learn, Biopython), R (Bioconductor, Tidymodels), SQL, BASH

Bioinformatics

Nextflow (DSL2), NGS Analysis, Multiple Sequence Alignment (MAFFT), Sequence Trimming (trimAI), NCBI / Entrez

Biostatistics & ML

Deep Learning Foundations (TensorFlow, PyTorch concepts via JAX), Multivariate Analysis, PCA, Clustering, Scikit-learn, Tidymodels

Data Engineering

Hardware Acceleration (TPU/GPU), Large dataset processing (HDF5, NetCDF), Pipeline development, R Shiny, devtools/roxygen2

Tools

Linux/Unix, Git, HPC clusters, Conda

TRAINING / COURSES

IBM: Machine Learning with Python; Introduction to AI

Coursera (in progress)

High Performance Computing (HPC) Training

Jülich Supercomputing Center (JSC), Forschungszentrum Jülich

- Finalised the computational modelling framework and prepared the simulation data for peer-reviewed publication.

Job Search & German Language Learning
Hamburg, Germany

02/2025 - 04/2025 Hamburg, Germany

Active job search and German language acquisition (Goethe B1 certification).

Post-doctoral Scientist
University of Hamburg, Institute of Marine Ecosystem and Fisheries Sciences

08/2021 - 01/2025 Hamburg, Germany

- Marine Ecosystem Modelling
- Elternzeit (Parental Leave) from 01/2024 to 12/2024.*
 - Model Development: Developed the novel "Cyanobacteria Life Cycle" (CLC) model within the ERGOM framework for climate warming projections.
 - High-Performance Computing & ML: Led the translation of a legacy Fortran continuous model into a high-performance Python simulation engine. Utilized **Google JAX** vectorization and branchless logic for massive parallelization (direct compatibility with **TPUs and GPUs**).
 - This architecture established a strong foundation in tensor operations underpinning deep learning frameworks like **PyTorch** and **TensorFlow**.

Job Search
Kerala, India

04/2021 - 07/2021 Kerala, India

Relocation from India to Germany and active job search.

Physics Educator
Tutorwaves Solutions Inc / Freelance

10/2018 - 03/2021 Kerala, India

Prepared physics content for competitive examinations and coached state-level chess teams.

Ph.D. Researcher in Astrochemistry
University of Paris-Saclay

10/2015 - 09/2018 Orsay, France

- Led atomic collision experiments, analyzed data, and modeled outcomes.
- Acquired and processed terabytes of raw data from tandem particle accelerator experiments.
 - C++ signal processing, algorithm development, and molecular fragmentation modeling (KIDA database).

EDUCATION

Ph.D. in Astrochemistry
Université Paris-Saclay, Institute of Molecular Sciences (ISMO) Orsay, France

09/2015 - 09/2018 Orsay, France

M.Sc. in Physics
National Institute of Technology Calicut, India

07/2012 - 01/2015 Calicut, India

B.Sc. in Physics
University of Calicut, India

06/2009 - 04/2012 Thrissur, India

REFERENCES

Dr. Andreas Busjahn
abusjahn@healthtwist.de

Prof. Dr. Kai Wirtz
kai.wirtz@hereon.de

Prof. Dr. Elisa Schaum
elisa.schaum@uni-hamburg.de

MITx: 7.00x Introduction to Biology - The Secret of Life

Comprehensive coursework on genetics, biochemistry, and molecular biology.

OTHER SELECTED PROJECTS

Hepatitis Delta Virus Pipeline • GitHub
Nextflow (DSL2) pipeline for viral sequence analysis: NCBI Entrez download, MAFFT alignment, trimAl trimming, Conda-managed dependencies.

JAX/TPU High-Performance Simulator • GitHub
Ported a legacy Fortran model to a modular Python architecture using Google JAX. Engineered branchless, vectorized logic for massive TPU/GPU parallelization.

Introduction to Tidymodels • GitHub
Teaching script for ML in R: recipes, parsnip, workflows, and yardstick on the Palmer Penguins dataset.

Berlin Tree Analysis • GitHub
Interactive Jupyter notebook analyzing 5 years of tree planting data in Berlin using Python, Pandas, and Seaborn.

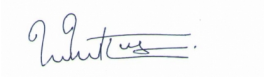
PUBLICATIONS

5 peer-reviewed articles • Google Scholar Profile

LANGUAGES

English	●●●●●●
Proficient	
German	●●●●●●
B1 - Goethe Certified	
French	●●●●●●
Intermediate	
Malayalam	●●●●●●
Native	
Tamil	●●●●●●
Proficient	
Hindi	●●●●●●
Intermediate	

Hamburg, 11 April 2026



Dr. Thejus Mahajan